

1. Given: $U_1, \dots, U_m$.

finite dim vector ~~spaces~~ ^{subspaces} of V s.t.

$U_1 + \dots + U_m$ is a direct sum.

To prove: $U_1 + \dots + U_m$ is finite dimensional &
 $\dim(U_1 \oplus \dots \oplus U_m) = \dim U_1 + \dots + \dim U_m$

Proof: i) ~~for~~ Let $B_i = \text{basis}(U_i)$
 $i \in [1, m]$

$\therefore \dim(U_i)$ is finite,
 B_i is a finite set

$\therefore$ Let $B = \bigcup_{i=1}^m B_i$

$\therefore B = B_1 \cup \dots \cup B_m$.

$\therefore B$ is finite set as it's a union of finite sets.

B spans $U_1 + \dots + U_m$ as

B_i spans U_i for $i \in [1, m]$

$\therefore U_1 + \dots + U_m$ is finite dimensional.

ii). Let

$$\sum_{i=1}^m \sum_{v \in B_i} a_v v = 0$$

by grouping terms (wrt subspaces)

$$v_1 + v_2 + \dots + v_m = 0, \\ v_i \in U_i.$$

As $\mathcal{B}$ is a DIRECT SUM,

$$\Rightarrow v_1 = v_2 = \dots = v_m = 0.$$

$\therefore a_v = 0$ for each $B_i = \text{basis}(U_i)$

$\therefore B$ is linearly independent.

$$\therefore \dim(U_1 \oplus \dots \oplus U_m) = |B|$$

$$= |B_1| + \dots + |B_2| = \dim(U_1) +$$

...

$$\dim(U_m)$$

2). $T(z_1, z_2, z_3) = (2z_1, 0, 5z_3)$

To find:

- all eigenvalues of T .
- Eigenspace corresponding to each eigenvalue.

a). Let $(z_1, z_2, z_3) \neq (0, 0, 0)$

Then, $(2z_1, 0, 5z_3) = (\lambda z_1 + \lambda z_2, \lambda z_3)$

$$\lambda z_2 = 0.$$

$$(\lambda - 5) z_3 = 0$$

$$\lambda z_1 = 2 z_2.$$

Now, Let: $z_3 = 0$, or $\lambda = 5$.

Let $z_2 = 0$ or $\lambda = 0$.

So: $\lambda = 5$.

$$z_2 = 0.$$

$$z_1 = 0$$

$$z_3 = x \therefore \text{eigenvalue} = \boxed{5}$$

2). $\lambda = 0$,

$$5z_1 = 0, \quad z_2 = 0$$

$$\therefore z_3 = 0$$

$$z_1 \neq 0$$

$$\therefore \text{eigenspace} = 0.$$

for ANY other λ .

$$z_1 = 0, \quad z_2 = 0, \quad z_3 = 0$$

$\therefore$ NOT an eigenspace.

Eigenspaces are 0 & 5.

b). for $\lambda = 0$,

$$E_0 = \{ (z, 0, 0) : z \in \mathbb{R} \}$$

$$\therefore \boxed{B_0 = \text{span} \{ (1, 0, 0) \}}$$

$$E_5 = \{ (0, 0, z_3) : z_3 \in \mathbb{F}_3 \} = \text{span}\{ (0, 0, 1) \}$$

$$\therefore E_5 = \text{span}\{ (0, 0, 1) \}$$

3). Given: V is a vector space.

$A \subseteq V$ be a set of linearly independent vectors.

$B \subseteq V$ be a set for linearly independent vectors.

$$|A| < |B|.$$

True or False: $\exists n, n \in B \setminus A$

such that $A \cup \{n\}$ is ALSO a linearly independent set.

Proof: Proof by contradiction:

Let's assume:

$\forall n \in B \setminus A$, $A \cup \{n\}$ is linearly dependent.

$$\therefore n \in \text{span}(A), \forall n \in B \setminus A.$$

$$\therefore \nexists b \in B,$$

$$b \in A \text{ OR } b \in \text{span}(A)$$

$$\therefore B \subseteq \text{span}(A)$$

As B is linearly independent

$$|B| \leq \dim(\text{span}(A)) = |A|$$

But, contradiction:

$$|A| < |B| \rightarrow \text{given.}$$

$\therefore$ The assumption is false.

The statement in the question is TRUE.

$\therefore \exists x \in B \setminus A$ s.t. $A \cup \{x\}$ is linearly independent.

4). $T \in L(V)$ & $S \in L(V)$ is invertible.

To prove: a). T & $S^{-1}TS$ have SAME eigenvalues.

b). Relation b/w eigenvectors of T and $S^{-1}TS$.

Proof:

a). Let λ be an eigenvalue of T with eigenvector $v \neq 0$.

$$Tv = \lambda v.$$

$$\text{Let } w = S^{-1}v.$$

as S is invertible, $w \neq 0$.

$$\begin{aligned} S^{-1}TS(w) &= S^{-1}T(Sw) = S^{-1}T(SS^{-1}v) \\ &= S^{-1}T(v) = S^{-1}(Tv) = S^{-1}(\lambda v) \\ &= \lambda(S^{-1}v) = \lambda w. \end{aligned}$$

$$S^{-1}TS(w) = \lambda w.$$

$\therefore \lambda$ is also an eigenvalue of S^{-1}

for opp direction,

$$v = Sw.$$

~~for~~ $S^{-1} T S(w) = \lambda w.$

$$T(v) = T(Sw)$$

$$S^{-1} T(v) = S^{-1} T S(w)$$

$$S^{-1} T(v) = \lambda w.$$

$$T(v) = S(\lambda w)$$

$$T(v) = \lambda(S(w))$$

$$\boxed{T(v) = \lambda w} \quad \therefore$$

b), if v is an eigenvector of T ,

$$Tv = \lambda v.$$

$$w = S^{-1}v$$

$$S^{-1}TS(w) = \lambda w.$$

$\therefore$ Eigenvectors of $S^{-1}TS$ are
of the form $\boxed{S^{-1}v}$

where v is an eigenvector
of T .

similarly,

~~w is~~ v is an eigenvector of
 T ,

$$\text{so } \boxed{v = Sw} \text{ where}$$

w is an eigenvector of
 $S^{-1}TS$.

5). Proof to prove:

$T \in L(V)$ has diagonal matrix w.r.t basis (V) , the diagonal entries are eigenvalues of T .

Proof:

Let $\beta = \{v_1, v_2, \dots, v_n\}$

matrix of T w.r.t β is diagonal.

$$[T]_{\beta} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

ith column of $[T]_{\beta}$ gives

co-ordinates of $T(v_i)$ in $\text{basis}(\beta)$

$\therefore$ it has λ_i in the position i is elsewhere.

$$\therefore T(v_i) = \lambda_i v_i$$

$$v_i \neq 0$$

$\therefore \lambda_i$ is eigenvalue for
eigenvector v_i .

Let λ be eigenvalue of T ,
 $v \neq 0$.

$$v = a_1 v_1 + \dots + a_n v_n$$

$\S$ Not all $a_i = 0$.

$\therefore$ There are no other eigenvalues

As:

$\Rightarrow$

$$T(v) = a_1 d_1 v_1 + \dots + a_n d_n v_n$$

$$T(v) = \lambda v = a_1 \lambda v_1 + \dots + a_n \lambda v_n$$

$$\therefore a_i d_i = a_i \lambda \quad \forall i$$

$$a_i (d_i - \lambda) = 0 \quad \forall i$$

as not all a_i are 0 as

$$v \neq 0,$$

$$\S \lambda = d_i.$$

$\therefore$ Every diagonal entry is an eigenvalue & every eigenvalue is equal to some d_i .

6). $P \in \mathbb{C}$, $T \in L(V)$; $p \in P(\mathbb{C})$ is a polynomial & $\alpha \in \mathbb{C}$.

To prove: α is eigenvalue of $p(T)$ iff $\alpha = p(\lambda)$ for some eigenvalue λ of T .

To Proof:

($\Rightarrow$) forward proof:

Assume $p(T)v = \alpha v$, $v \neq 0$.

$$\therefore q(z) = p(z) - \alpha$$

$$q(T)(v) = (p(T) - \alpha I)v = 0.$$

Now,

$$q(z) = c(z - \lambda_1)(z - \lambda_2) \dots (z - \lambda_n)$$

$$\therefore q(T) = c(T - \lambda_1 I) \dots (T - \lambda_n I)$$

$$\text{Now, } (T - \lambda_1 I) \dots (T - \lambda_n I)v = 0$$

$\therefore$ Proof by contradiction:

If all factors $(T - \lambda_i I)$ were invertible, their product would be invertible.

But $v \neq 0$.

$\therefore$ for some λ ,

$T - \lambda I$ is NOT invertible

$\therefore \lambda$ is eigenvalue of T .

Since $f(\lambda) = 0$,

$$p(\lambda) = \lambda \neq 0$$

$\lambda = p(\lambda)$

Backward proof : ($\Leftarrow$)

$Tv = \lambda v$ for some $v \neq 0$,

$$p(z) = a_0 + a_1 z + \dots + a_n z^n,$$

$$\therefore p(T)(v) = a_0 v + a_1 T v + \dots + a_n T^n v.$$

using eigenvector property:

$$T^k v = \lambda^k v,$$

$$\begin{aligned} p(T)(v) &= (a_0 + a_1 \lambda + \dots + a_n \lambda^n) v \\ &= p(\lambda) v. \end{aligned}$$

$$\text{Let } p(\lambda) = \alpha,$$

$$\therefore p(T)v = \alpha v.$$

$\therefore \alpha$ is eigenvalue of $p(T)$ iff
 $\alpha = p(\lambda)$ for some λ of T

eigenvalue